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Ladder and ladder top

Abstract

A ladder having a first rail. The ladder having a second rail. The ladder having a plastic top directly attached to the first rail and second rail with fasteners. The top having a first side and a second side. The first side having a gap with a surface with a first rectangular slot and a second rectangular slot extending from the first rectangular slot so an external corner can fit into the first rectangular slot when the top leans against the corner. The surface of the first side having a first notch in the gap so a first rectangular plank can fit into the first notch when the top leans against the plank. The first notch having a width slightly larger than a width of the first plank. A ladder top. A method for using a ladder.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. patent application Ser. No. 15/803,023 filed Nov. 3, 2017, now U.S. Pat. No. 10,590,703, which is a nonprovisional application of U.S. provisional application Ser. No. 62/520,935 filed Jun. 16, 2017, and is a continuation-in-part of U.S. patent application Ser. No. 15/395,397 filed Dec. 30, 2016 and U.S. design patent application Ser. No. 29/589,376 filed Dec. 30, 2016, now U.S. Pat. No. D835,807 issued Dec. 11, 2018, all of which are incorporated by reference herein.

FIELD OF THE INVENTION

(1) The present invention is related to a top of a ladder that can stably lean against an external corner or a rectangular plank whose width is less than the width of the top. (As used herein, references to the “present invention” or “invention” relate to exemplary embodiments and not necessarily to every embodiment encompassed by the appended claims.) More specifically, the present invention is related to a top of a ladder that can stably lean against an external corner or a rectangular plank whose width is less than the width of the top where the top has a first side with a surface and a gap, and the surface in the gap has a V or an upside down bottle top shape to conform to the corner when the top leans against a corner, and the surface in the gap has a notch to conform to the rectangular plank when the top leans against the rectangular plank.

BACKGROUND OF THE INVENTION

(2) This section is intended to introduce the reader to various aspects of the art that may be related to various aspects of the present invention. The following discussion is intended to provide information to facilitate a better understanding of the present invention. Accordingly, it should be understood that statements in the following discussion are to be read in this light, and not as admissions of prior art.

(3) Ladders are commonly used to lean against surfaces that are wider than the width of a ladder top, or against corners. However, there are other surfaces, such as rectangular planks whose width is less than the width of a ladder top. It is desirable to be able to stably lean a ladder top against rectangular planks whose width is less than the width of the ladder top as well as external and internal corners.

BRIEF SUMMARY OF THE INVENTION

(4) The present invention pertains to a ladder. The ladder comprises a first rail. The ladder comprises a second rail. The ladder comprises a plastic top directly attached to the first rail and second rail with fasteners. The top having a first side and a second side. The first side having a gap with a surface with a V so a corner can fit into the V when the top leans against the corner. The surface of the first side having a first notch in the gap so a first rectangular plank can fit into the first notch when the top leans against the plank. The first notch having a width slightly larger than a width of the first plank. The top having a first cushion bonded to the first side without any mechanical fasteners.

(5) The present invention pertains to a top for a ladder to be leaned against a first rectangular plank or an external corner. The top has a first rail and a second rail. The top comprises a middle portion having a perimeter with a first side, second side, third side and fourth side extending down from the perimeter. The first side has a gap with a surface with a V so a corner can fit into the V when the top leans against the corner. The surface of the first side having a first notch in the gap so a first rectangular plank can fit into the first notch when the top leans against the plank. The first notch has a width slightly larger than a width of the first plank. The top being one piece and made of plastic. The third side having a first fastener hole to receive a first fastener to permanently attach the first rail to the third side. The fourth side has a second fastener hole to receive a second fastener to permanently attach the second rail to the fourth side.

(6) The present invention pertains to a method for using a ladder. The method comprises the steps of moving the ladder to a first rectangular plank. There is the step of leaning the ladder against the first rectangular plank so the rectangular plank is disposed in a first notch in a gap in a first side of

a plastic top directly attached to a first rail and a second rail with fasteners. The top has a second side. The first side has the gap which has a surface with a V so a corner can fit into the V when the top leans against the corner. The surface of the first side having the first notch in the gap. The first notch having a width slightly larger than a width of the first plank.

(7) The present invention pertains to a ladder. The ladder comprises a first rail. The ladder comprises a second rail. The ladder comprises a plastic top directly attached to the first rail and second rail with fasteners. The top having a first side and a second side. The first side having a gap with a surface with a first rectangular slot and a second rectangular slot extending from the first rectangular slot so an external corner can fit into the first and second rectangular slots when the top leans against the corner. The surface of the first side having a first notch in the gap so a first rectangular plank can fit into the first notch when the top leans against the plank. The first notch having a width slightly larger than a width of the first plank. The top having a first cushion bonded to the first side without any mechanical fasteners.

(8) The present invention pertains to a top for a ladder to be leaned against a first rectangular plank or an external corner. The top is directly attached to a first rail and a second rail. The top comprises a middle portion having a perimeter with a first side, second side, third side and fourth side extending down from the perimeter. The first side has a gap with a surface with a first rectangular slot and a second rectangular slot extending from the first rectangular slot so an external corner can fit into the first and second rectangular slots when the top leans against the corner. The surface of the first side having a first notch in the gap so a first rectangular plank can fit into the first notch when the top leans against the plank. The first notch has a width slightly larger than a width of the first plank. The top being one piece and made of plastic. The third side having a first fastener hole to receive a first fastener to permanently attach the first rail to the third side. The fourth side has a second fastener hole to receive a second fastener to permanently attach the second rail to the fourth side.

(9) The present invention pertains to a method for using a ladder. The method comprises the steps of moving the ladder to a first rectangular plank. There is the step of leaning the ladder against the first rectangular plank so the rectangular plank is disposed in a first notch in a gap in a first side of a plastic top directly attached to a first rail and a second rail with fasteners. The top has a second side. The first side has the gap which has a surface with a first rectangular slot and a second rectangular slot extending from the first rectangular slot so an external corner, as shown in FIG. 29, can fit into the first and second rectangular slots when the top leans against the corner. The surface of the first side having the first notch in the gap. The first notch having a width slightly larger than a width of the first plank.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING

(1) In the accompanying drawings, the preferred embodiment of the invention and preferred methods of practicing the invention are illustrated in which:

(2) FIG. 1 shows an overhead view of a ladder top of the present invention.

(3) FIG. 2 shows a ladder with the ladder top.

(4) FIG. 3 shows an underside view of the ladder top.

(5) FIG. 4 shows a front view of the ladder top.

(6) FIG. 5 shows an overhead view of the ladder top against an external corner.

(7) FIG. 6 shows an overhead view of the ladder top against a first rectangular plank.

(8) FIG. 7 shows an overhead view of the ladder top against a second rectangular plank.

(9) FIG. 8 is an overhead view of the ladder top against an internal corner.

(10) FIG. 9 is an overhead view of the ladder top against a wall.

- (11) FIG. **10** is a perspective view of a stepladder of the present invention.
- (12) FIG. **11** is a perspective view of an alternative embodiment of the ladder top.
- (13) FIG. **12** is a top view of the alternative embodiment of the ladder top.
- (14) FIG. **13** is a left side view of the alternative embodiment of the ladder top.
- (15) FIG. **14** is a bottom view of the alternative embodiment of the ladder top.
- (16) FIG. **15** is a bottom view of a second alternative embodiment of the ladder top.
- (17) FIG. **16** is a front view of the second alternative embodiment of the ladder top.
- (18) FIG. **17** is a right side view of the second alternative embodiment of the ladder top.
- (19) FIG. **18** is a left side view of the second alternative embodiment of the ladder top.
- (20) FIG. **19** is an overhead view of the second alternative embodiment of the ladder top.
- (21) FIG. **20** is a rear view of the second alternative embodiment of the ladder top.
- (22) FIG. **21** shows a front perspective view of the second alternative embodiment of the ladder top.
- (23) FIG. **22** shows a rear perspective view of the second alternative embodiment of the ladder top.
- (24) FIG. **23** is an overhead view of an alternative embodiment of a top.
- (25) FIG. **24** is a rear view of the top of FIG. **23**.
- (26) FIG. **25** is an underside view of the top of FIG. **23**.
- (27) FIG. **26** is a front perspective view of the top of FIG. **23**.
- (28) FIG. **27** is a rear perspective view of the top of FIG. **23**.
- (29) FIG. **28** shows the top of FIG. **23** with a 2×4 in the first notch.
- (30) FIG. **29** shows the top of FIG. **23** with an external corner in the gap.
- (31) FIG. **30** is a perspective view of a step ladder with the top of FIG. **23**.
- (32) FIG. **31** is a partially exploded view of the top and the cushions.
- (33) FIG. **32** shows preferred dimensions of aspects of the gap of the top.

DETAILED DESCRIPTION OF THE INVENTION

(34) Referring now to the drawings wherein like reference numerals refer to similar or identical parts throughout the several views, and more specifically to FIGS. **1-4** and **10**, thereof, there is shown a ladder **10**. The ladder **10** comprises a first rail **12**. The ladder **10** comprises a second rail **14**. The ladder **10** comprises a plastic top **16** directly attached to the first rail **12** and second rail **14** with fasteners **78**. The top **16** has a first side **18** and a second side **20**. The first side **18** has a gap **22** with a surface **24** with a V **26** so an external corner **28**, as shown in FIG. **5**, can fit into the V **26** when the top **16** leans against the corner **28**. The surface **24** of the first side **18** has a first notch **30** in the gap **22** so a first rectangular plank **34** can fit into the first notch **30** when the top **16** leans against the plank, as shown in FIG. **6**. The first notch **30** has a width slightly larger than a width of the first plank **34**. The top **16** having a first cushion **70** bonded to the first side without any mechanical fasteners. As shown in FIG. **14**, the cushion **70** may be connected via barbs or protrusions **134** extending through cavities or openings **135**.

(35) The first notch **30** may be in the V **26**. The surface **24** of the first side **18** may have a second notch **32** in the gap **22** having a width greater than the width of the first notch **30** so a second rectangular plank **36** having a width slightly smaller than the width of the second notch **32** can fit into the second notch **32** when the top **16** leans against the second plank **36**, as shown in FIG. **7**. The second notch **32** may be in the V **26**.

(36) The surface **24** of the first side **18** may have a first straight segment **38**, a second straight segment **40**, a third straight segment **42**, a fourth straight segment **44**, a fifth straight segment **46**, a sixth straight segment **48**, a seventh straight segment **50** and an eighth straight segment **52**, as shown in FIG. **1**. The second straight segment **40** may directly connect with the first straight segment **38** to form a bottom **88** of the V **26**. The third straight segment **42** may directly connect with the first straight segment **38** at an angle greater than 100° and less than 170°. The fourth straight segment **44** may directly connect with the second straight segment **40** at an angle greater than 100° and less than 170°. The fifth straight segment **46** may directly connect with the third

straight segment **42** at an angle of about 90° . The sixth straight segment **48** may directly connect with the fourth straight segment **44** at an angle of about 90° . The third, fourth, fifth and sixth straight segments **42**, **44**, **46**, **48** together forming the first notch **30**.

(37) The surface **24** of the first side **18** may have a ninth straight segment **54**, a tenth straight segment **56**, an eleventh straight segment **58** and a twelfth straight segment **60**. The eleventh straight segment **58** may directly connect with the ninth straight segment **54** at an angle of about 90° . The twelfth straight segment **60** may directly connect with the tenth straight segment **56** at an angle of about 90° . The ninth, tenth, eleventh and twelfth straight segments **54**, **56**, **58**, **60** together forming the second notch **32** in the gap **22** having a width greater than the width of the first notch **30** so the second rectangular plank **36** having a width slightly smaller than the width of the second notch **32** can fit into the second notch **32** when the top **16** leans against the second plank **36**.

(38) The surface **24** of the first side **18** may have a thirteenth straight segment **62** and a fourteenth straight segment **64**. The thirteenth straight segment **62** may directly connect with the fifth straight segment **46** and the ninth straight segment **54** and is in linear alignment with the first straight segment **38**. The fourteenth straight segment **64** may directly connect with the sixth straight segment **48** and the tenth straight segment **56** and is in linear alignment with the second straight segment **40**.

(39) The first cushion **70** may be attached to the surface **24** of the first side **18** in the gap **22** at the thirteenth straight segment **62** and a second cushion **72** attached to the surface **24** of the first side **18** in the gap **22** at the fourteenth straight segment **64**. The first and second cushions **70**, **72** protect the corner **28** from being scratched or marred by the top **16** when the top **16** leans against the corner **28**. The top **16** may have a third side **84** directly connected to the first side **18** and the second side **20** and a fourth side **86** directly connected to the first side **18** and the second side **20**. The first rail **12** may be permanently attached with a first fastener **80** to the third side **84** and the second rail **14** maybe permanently attached with a second fastener **82**. The ladder **10** may include a third cushion **74** attached to the surface **24** of the first side **18** at the seventh straight segment **50** and a fourth cushion **76** attached to the surface **24** of the first side **18** at the eighth straight segment **52**.

(40) The top **16** may have a middle portion **92** disposed between a perimeter **94** of the top **16** defined by the first, second, third and fourth sides **18**, **20**, **84**, **86**. The middle portion **92** may have a paint can indentation **96** to hold a paint can, and a tool hole **90** to hold a tool, such as a screwdriver or pliers or drill. There may be several tool holes **90** in the middle portion **92** to accommodate a number of different types of tools.

(41) The surface **24** of the first side **18** may have a fifteenth straight segment **66** that directly connects with the third side **84** and to the seventh straight segment **50** at an angle between 20° and 70° , and a sixteenth straight segment **68** that directly connects with the fourth side **86** and the eighth straight segment **52** at an angle between 20° and 70° . As shown in FIG. **8**, the angular position of the seventh and eighth straight segments allow the seventh and eighth straight segments, and thus the top **16**, to fit stably against an internal corner **98** when the top **16** leans against an internal corner **98**. Additional cushions may be used as desired along the first side **18**. When leaning against a wide surface, such as a wall **100** as shown in FIG. **9**, the third cushion on the seventh straight segment and the fourth cushion on the eighth straight segment of the top **16** will rest against the wall **100**.

(42) The present invention pertains to a top **16** for a ladder **10** to be leaned against a first rectangular plank **34** or an external corner **28**. The top has a first rail and a second rail attached to it. The top comprises a middle portion **92** having a perimeter **94** with a first side **18**, second side **20**, third side **84** and fourth side **86** extending down from the perimeter **94**. The first side **18** has a gap **22** with a surface **24** with a V **26** so a corner **28** can fit into the V **26** when the top **16** leans against the corner **28**. The surface **24** of the first side **18** having a first notch **30** in the gap **22** so a first rectangular plank **34** can fit into the first notch **30** when the top **16** leans against the plank **34**. The first notch **30** has a width slightly larger than a width of the first plank **34**. The top **16** being one

piece and made of plastic and the cushions made of rubber. The third side **84** having a first fastener hole **102** to receive a first fastener **80** to permanently attach the first rail **12** to the third side **84**. The fourth side **86** has a second fastener hole **104** to receive a second fastener **82** to permanently attach the second rail **14** to the fourth side **86**.

(43) The present invention pertains to a method for using a ladder **10**. The method comprises the steps of moving the ladder **10** to a first rectangular plank **34**. There is the step of leaning the ladder **10** against the first rectangular plank **34** so the rectangular plank **34** is disposed in a first notch **30** in a gap **22** in a first side **18** of a plastic top **16** directly attached to a first rail **12** and a second rail **14** with fasteners **78**. The top **16** has a second side **20**. The first side **18** has the gap **22** which has a surface **24** with a V **26** so a corner **28** can fit into the V **26** when the top **16** leans against the corner **28**. The surface **24** of the first side **18** having the first notch **30** in the gap **22**. The first notch **30** having a width slightly larger than a width of the first plank **34**.

(44) In the operation of the invention, in one embodiment, a molded top **16** fits on the upper rail ends of the fly section of an extension ladder **10**. The top **16** has locations at its middle portion **92** to permit the temporary placement of tools when the user is working at the upper end of the ladder **10**. The back or first side **18** of the top **16** is specifically designed so that the ladder **10** may be leaned against a wide flat surface, such as a wall **100** as shown in FIG. **9**, may be leaned against an external corner **28** as shown in FIG. **5**, and may be leaned against and closely engage a rectangular plank, such as 6" wide or 4" wide lumber as shown in FIGS. **6** and **7**.

(45) When leaning against an external corner **28**, the V shape of the external corner **28** fits into the V **26** of the top **16** formed by the first straight segment **38** and the second straight segment **40**. The first straight segment **38** directly connects with the second straight segment **40**, typically at an angle of about 90°, to match a typical external corner **28**. In addition, the external corner **28**, when the top **16** is leaning against the corner **28** and the external corner **28** is in the gap **22**, the external corner **28** also contacts the first cushion **70** on the thirteenth straight segment **62**, which is in linear alignment with the first straight segment **38**, and the second cushion **72** which is on the fourteenth straight segment **64** which is in linear alignment with the second straight segment **40**. By being in linear alignment with the first and second straight segments **38**, **40**, the thirteenth straight segment **62** and the fourteenth straight segment **64**, respectively, effectively extend the length of the V **26** to provide a larger V and thus a larger surface area for the top **16** to rest against and consequently a more secure and stable fit of the top **16** with the external corner **28**.

(46) Similarly, when the top **16** leans against an internal corner **98**, the fifteenth straight segment **66** and the sixteenth straight segment **68**, which directly connect to and form an angle of about 45° with the seventh straight segment **50** and the eighth straight segment **52**, respectively, stably rest against and align with the walls that form the internal corner **98**. In this way the top can be disposed in an internal corner for a user to use, with a greater surface area of the top **16** contacting the walls of the internal corner **98** than if the fifteenth and sixteenth straight segments did not exist or were not that about a 45° angle relative to the seventh and eighth straight segments.

(47) The first notch **30** and the second notch **32** are disposed in, or integrated with and break up the V **26** in the gap **22**. The first notch **30** conforms to the rectangular cross-section of a rectangular plank, such as a 4 inch wide plank. The first notch **30** is slightly larger in width than the width of the 4 inch plank so that when the top **16** leans against the 4 inch plank, the first notch **30** receives the 4 inch plank rather snugly so there is essentially no movement between the top **16** and the 4 inch plank and the top **16** is stably positioned with the 4 inch plank.

(48) Similarly, the second notch **32** conforms to the rectangular cross-section of a rectangular plank, such as a 6 inch wide plank. The second notch **32** is slightly larger in width than the width of the 6 inch plank so that when the top **16** leans against the 6 inch plank, the second notch **32** receives the 6 inch plank rather snugly so there is essentially no movement between the top **16** and the 6 inch plank top **16** is stably positioned with the 6 inch plank.

(49) Outside the gap **22** on either side of the top **16** there is a third cushion **74** and forth cushion

which will contact a flat section, such as a wall, when the top **16** leans against the wall. The third and fourth cushions **74**, **76** protect the wall from being marred or scratched by the top **16** when the top **16** leans against the wall.

(50) The ladder top **16** with the cushions or pads is formed from 2 distinct materials. A mold utilizing a rotating platen to process two materials in the same part produces the ladder top **16** with the cushions. The ladder top **16** is made of polypropylene and the soft rubber pads or cushions are made of a ThermoPlastic Vulcanizate (TPV), specifically under the brand name Santoprene (TPV). In the production process, a mold for the top **16** receives the molten plastic. A removable portion of the mold creates cavities or openings **135** in the first side **18** where the cushions will ultimately be located. After the top **16** is formed, it stays in the stationary portion of the mold and the removable portion of the mold is replaced with a second removable portion of the mold. The overall mold is then closed and liquid TPV is introduced into the mold to fill the cavities that are then present in the preformed top **16**. The TPV is allowed to cool resulting in the presence of the cushions with the top **16**. These cushions form both mechanical and a chemical bond with the top **16**, for added strength to prevent the cushions from separating from the first side **18**. The molten TPV chemically reacts with the polypropylene to create the chemical bond, as well as with the cooling TPV forming a mechanical bond. The top **16** may be one continuous piece, including the cushions, given the cushions are chemically bonded to the segments.

(51) The rails and the rungs themselves are standard commonly available components. The width of the top **16** is preferably between 13 and 20 inches. The length of the ladder top is between 5 and 10 inches. The height of the ladder top is between 2 and 7 inches. The thickness of the ladder top is between $\frac{1}{8}$ of an inch and $\frac{3}{8}$ of an inch. Buttressing (each rib being between 0.1 and 0.2 inches thick) underneath the top **16** can be utilized as shown in the figures. The ladder **10** with any of the embodiments described herein of the top **16** when leaning against a surface at 75.5° can support a load on the ladder of at least 600 pounds without failing.

(52) Referring to FIGS. **23-27** and **30**, thereof, there is shown a ladder **10**. The ladder **10** comprises a first rail **12**. The ladder **10** comprises a second rail **14**. The ladder **10** comprises a plastic top **16** directly attached to the first rail **12** and second rail **14** with fasteners **78**. The top **16** has a first side **18** and a second side **20**. The first side **18** has a gap **22** with a surface **24** with a first rectangular slot **27** and a second rectangular slot **29** extending from the first rectangular slot **27**, essentially forming an upside down bottle top shape, so an external corner **28**, as shown in FIG. **29** can fit into the first and second rectangular slots **27**, **29** when the top **16** leans against the corner **28**. The surface **24** of the first side **18** has a first notch **30** in the gap **22** formed from the first and second rectangular slots so a first rectangular plank **34** can fit into the first notch **30** when the top **16** leans against the plank **34**, as shown in FIG. **28**. The first notch **30** has a width slightly larger than a width of the first plank **34**. The top **16** having a first cushion **70** bonded to the first side without any mechanical fasteners.

(53) The surface **24** of the first side **18** may have a second notch **32** in the gap **22** having a width greater than the width of the first notch **30** so a second rectangular plank **36** having a width slightly smaller than the width of the second notch **32** can fit into the second notch **32** when the top **16** leans against the second plank **36**, as shown in FIG. **7**. The second notch **32** may be in the gap **22** disposed in front of the first notch **30**.

(54) The surface **24** of the first side **18** may have a first straight segment **38**, a second straight segment **40**, a third straight segment **42**, a fourth straight segment **44**, a fifth straight segment **46**, a sixth straight segment **48**, a seventh straight segment **50** and an eighth straight segment **52** and a bottom straight segment **88**. Segments **38**, **40**, **46**, **48** are perpendicularly positioned relative to the second side **20**. Segments **88**, **42**, **44**, **50** and **52** are horizontally positioned relative to the second side **20**, as shown in FIG. **23**. The second straight segment **40** and the first straight segment **38** may directly connect with the bottom segment **88** of the gap **22**. The third straight segment **42** may directly connect with the first straight segment **38** to form a right angle. The fourth straight segment **44** may directly connect with the second straight segment **40** and form a right angle. The fifth

straight segment **46** may directly connect with the third straight segment **42** and form a right angle. The sixth straight segment **48** may directly connect with the fourth straight segment **44** and form a right angle. The third, fourth, fifth and sixth straight segments **42, 44, 46, 48** together forming the first notch **30**.

(55) The surface **24** of the first side **18** may have a ninth straight segment **54**, a tenth straight segment **56**, an eleventh straight segment **58** and a twelfth straight segment **60**. The eleventh straight segment **58** may directly connect with the ninth straight segment **54** at an angle of about 90° . The twelfth straight segment **60** may directly connect with the tenth straight segment **56** at an angle of about 90° . The ninth, tenth, eleventh and twelfth straight segments **54, 56, 58, 60** together forming the second notch **32** in the gap **22** having a width greater than the width of the first notch **30** so the second rectangular plank **36** having a width slightly smaller than the width of the second notch **32** can fit into the second notch **32** when the top **16** leans against the second plank **36**.

(56) The surface **24** of the first side **18** may have a thirteenth straight segment **62** and a fourteenth straight segment **64**. The thirteenth straight segment **62** may directly connect with the fifth straight segment **46** and connect with the ninth straight segment **54** at an angle of between 20° and 70° and preferably about 45° . The fourteenth straight segment **64** may directly connect with the sixth straight segment **48** and connect with the tenth straight segment **56** at an angle of between 20° and 70° and preferably about 45° . An external corner rests against the straight segments **62, 64** when the top **16** leans against the corner, as shown in FIG. **29**, with the corner extending into the first rectangular slot **27** and the second rectangular slot **29**, if necessary.

(57) The first cushion **70** may be attached to the surface **24** of the first side **18** in the gap **22** at the thirteenth straight segment **62** and a second cushion **72** attached to the surface **24** of the first side **18** in the gap **22** at the fourteenth straight segment **64**. The first and second cushions **70, 72** protect the corner **28** from being scratched or marred by the top **16** when the top **16** leans against the corner **28**. The top **16** may have a third side **84** directly connected to the first side **18** and the second side **20** and a fourth side **86** directly connected to the first side **18** and the second side **20** and in parallel and spaced relationship with the third side **84**. The first rail **12** may be permanently attached with a first fastener **80** to the third side **84**, as shown in FIG. **27**, and the second rail **14** maybe permanently attached with a second fastener **82**, as shown in FIG. **26**, to the fourth side **86**. The ladder **10** may include a third cushion **74** attached to the surface **24** of the first side **18** at the seventh straight segment **50** and a fourth cushion **76** attached to the surface **24** of the first side **18** at the eighth straight segment **52**.

(58) The top **16** may have a middle portion **92** disposed between a perimeter **94** of the top **16** defined by the first, second, third and fourth sides **18, 20, 84, 86**. The middle portion **92** may have a paint can indentation **96** and a second larger paint can indentation **97** to hold a paint can, a ramp **111** extending from the larger paint can indentation **97** for a user to more easily scoop up small parts such as screws or bolts that may have been put into the indentations, and a tool hole **90** to hold a tool, such as a screwdriver or pliers or drill. There may be several tool holes **90** in the middle portion **92** to accommodate a number of different types of tools.

(59) The surface **24** of the first side **18** may have a fifteenth straight segment **66** that directly connects with the third side **84** and to the seventh straight segment **50** at an angle between 20° and 70° , and a sixteenth straight segment **68** that directly connects with the fourth side **86** and the eighth straight segment **52** at an angle between 20° and 70° . As shown in FIG. **8**, the angular position of the seventh and eighth straight segments allow the seventh and eighth straight segments, and thus the top **16**, to fit stably against an internal corner **98** when the top **16** leans against an internal corner **98**. Additional cushions may be used as desired along the first side **18**. When leaning against a wide surface, such as a wall **100** as shown in FIG. **9**, the third cushion on the seventh straight segment and the fourth cushion on the eighth straight segment of the top **16** will rest against the wall **100**. The FIGS. **6, 7, 8** and **9** show the embodiment of the top **16** with the V shape, but this top **16** with the rectangular slots, as shown in FIG. **23**, could be used in the same

way as depicted in FIGS. 6, 7, 8 and 9.

(60) The ladder **10** has a plurality of steps **11** attached to the first and second rails forming a front assembly **21**, and a plurality of cross bars **13** attached to a third rail **154** and a fourth rail **156** forming a rear assembly **27**, as shown in FIGS. **10** and **30**.

(61) The present invention pertains to a top **16** for a ladder **10** to be leaned against a first rectangular plank **34** or an external corner **28**. The top **16**, as shown in FIGS. **23** and **30**, is attached to a first rail **12** and a second rail **14**. The top **16** comprises a middle portion **92** having a perimeter **94** with a first side **18**, second side **20**, third side **84** and fourth side **86** extending down from the perimeter **94**. The first side **18** has a gap **22** with a surface **24** with a first rectangular slot **27** and a second rectangular slot **29** extending from the first rectangular slot **27** so an external corner **28**, as shown in FIG. **29** can fit into the first and second rectangular slots **27**, **29** when the top **16** leans against the corner **28**. The surface **24** of the first side **18** having a first notch **30** in the gap **22** so a first rectangular plank **34** can fit into the first notch **30** when the top **16** leans against the plank **34**, as shown in FIG. **28**. The first notch **30** has a width slightly larger than a width of the first plank **34**. The top **16** being one piece and made of plastic and the cushions made of rubber. The third side **84** having a first fastener hole **102** to receive a first fastener **80** to permanently attach the first rail **12** to the third side **84**, as shown in FIG. **27**. The fourth side **86** has a second fastener hole **104** to receive a second fastener **82** to permanently attach the second rail **14** to the fourth side **86**, as shown in FIG. **26**.

(62) As shown in FIG. **2** and FIG. **4**, in this case for an extension ladder, the top **16** has a first flap **120** having a rectangular shape which conforms with the shape of the web of the top side of the first rail **12**, and having a first fastener hole **102** that receives a first fastener **78** to fasten the first flap **120** to the web of the top side of the first rail **12**. The first flap **120** extends down from the middle portion **92** of the top **16** near to and inside of the third side **84**. The third side **84** has an L shape with a flat portion **122** that lines with and turns into the first side **18**, and a lower portion **124** that extends down from the flat portion **122**. The second side **20** extends down from the perimeter **94** the same distance as the lower portion extends down from the perimeter and forms a smooth continuous surface with the lower portion. In the space created between the first flap **120** and the lower portion **124** the top side of the first rail **12** fits and is essentially protected somewhat on the outside from the lower portion **124** and at the front from the second side **20** and from the inside by the first flap **120**. The second rail **14** is similarly attached to a second flap **130** in spaced relationship with the fourth side **86**. In this manner, the top **16** attaches to the first and second rails of an extension ladder. It should be noted, the C shape of the rails face outward since they engage with rails of the lower section of the extension ladder whose cross-sectional shape faces inwards. The flaps are necessary to provide a flat mounting surface to the web of the rails of the top section of the extension ladder since only the edges of the flanges of the rails contact the third and fourth sides in the extension ladder embodiment.

(63) In this manner, the ladder top **16** can also be attached to a step ladder **140**, as shown in FIG. **10**, where the stepladder has a first hinge **150** that attaches to and extends from the first rail **12** and a second hinge **152** that attaches to and extends from the second rail **14**. The first and second hinges are in spaced relation with the ladder top **16**. A third rail **154** attaches to the first hinge **150** and a fourth rail **156** attaches to the second hinge **152**. The third rail **154** and the fourth rail **156** are able to move between an open, use position, and a closed position with the first hinge **150** and second hinge **152**, respectively, as is well known in the art. There are rungs or cross bars **13** attached to the third and fourth rails, as well as brackets which fold attached between the first rail **12** and the third rail **154**, and between the second rail **14** and the fourth rail **156**. When the stepladder is in a closed position, with the rails essentially in parallel with each other, the stepladder can be leaned on the top **16** against structures, such as described above, with the notches for planks and the first and second rectangular slots for an external corner and the angled corner segments for an internal corner, and be more securely kept in position without sliding or falling

over.

(64) FIGS. **11-14** and **23-29** show additional embodiment of the top **16** that is preferably used with the stepladder embodiment. Note, there are no flaps, but instead, the first rail **12** attaches to the third side **84** with fasteners **78** extending through fastener holes in the third side **84** and the first rail **12**; and the second rail **14** attaches to the fourth side **86** with fasteners **78** extending through fastener holes in the fourth side **86**, and the second rail **14**. In the stepladder embodiment, the C shape cross section of the rails face inward, so the web of the rails contact directly the third and fourth sides of the top and no flaps are needed. See U.S. patent application Ser. No. 15/802,949 having docket number WERNER-153 CIP, contemporaneously filed with this application, and incorporated by reference herein, for additional details regarding a stepladder.

(65) FIGS. **15-22** show a second alternative embodiment of the top **16** having bungee slots **180** in the third side **84** and fourth side **86** for holding tools with bungee straps. In addition, a magnet **170** is completely encapsulated in the top **16**, as shown in FIGS. **15** and **19**. When the top **16** is being formed, the magnet, which is a single solid element, is placed on pins in the mold so that when the molten plastic fills the mold it encompasses the magnet. The top shown in FIGS. **23-29** can also be used on an extension ladder and include flaps **120** and **130**, as described above. When used with a stepladder, since the C shaped cross-section of the rails face inward, the web of the rails are then positioned flat against the respective third and fourth sides, so there is a wide mounting surface to fix the rails to the third and fourth sides of the top, and no flaps are needed. Again, in an extension ladder, the C shaped cross section of the rails face outward, so the flaps are needed to fix the rails to the top.

(66) The present invention pertains to a method for using a ladder **10**. The method comprises the steps of moving the ladder **10** to a first rectangular plank **34**. There is the step of leaning the ladder **10** against the first rectangular plank **34** so the rectangular plank **34** is disposed in a first notch **30** in a gap **22** in a first side **18** of a plastic top **16** directly attached to a first rail **12** and a second rail **14** with fasteners **78**, as shown in FIGS. **23** and **30**. The top **16** has a second side **20**. The first side **18** has the gap **22** which has a surface **24** with a first rectangular slot **27** and a second rectangular slot **29** extending from the first rectangular slot **27** so an external corner **28**, as shown in FIG. **29** can fit into the first and second rectangular slots **27**, **29** when the top **16** leans against the corner **28**. The surface **24** of the first side **18** having the first notch **30** in the gap **22**. The first notch **30** having a width slightly larger than a width of the first plank **34**.

(67) In the operation of the invention, in one embodiment, a molded top **16** fits on the upper rail ends of the fly section of an extension ladder **10**. The top **16** has locations at its middle portion **92** to permit the temporary placement of tools when the user is working at the upper end of the ladder **10**. The back or first side **18** of the top **16** is specifically designed so that the ladder **10** may be leaned against a wide flat surface, such as a wall **100** as shown in FIG. **9**, may be leaned against an external corner **28** as shown in FIG. **29**, and may be leaned against and closely engage a rectangular plank, such as 6" wide or 2" wide lumber as shown in FIGS. **6** and **28**.

(68) When leaning against an external corner **28**, the V shape of the external corner **28** fits into the first and second slots of the top **16** formed by the first straight segment **38** and the second straight segment **40** and the bottom straight segment **88**.

(69) Similarly, when the top **16** leans against an internal corner **98**, the fifteenth straight segment **66** and the sixteenth straight segment **68**, which directly connect to and form an angle of about 45° with the seventh straight segment **50** and the eighth straight segment **52**, respectively, stably rest against and align with the walls that form the internal corner **98**. In this way the top can be disposed in an internal corner for a user to use, with a greater surface area of the top **16** contacting the walls of the internal corner **98** than if the fifteenth and sixteenth straight segments did not exist or were not that about a 45° angle relative to the seventh and eighth straight segments.

(70) Outside the gap **22** on either side of the top **16** there is a third cushion **74** and fourth cushion which will contact a flat section, such as a wall, when the top **16** leans against the wall. The third

and fourth cushions **74**, **76** protect the wall from being marred or scratched by the top **16** when the top **16** leans against the wall.

(71) FIG. **31** shows an alternative embodiment of attaching the cushions to the top. The top, when formed, has inserts **85** which conform to the cushions that will fit into the inserts **85**. The cushions have barbs **87** which fit through barb holes **89** to hold the cushions in place in the inserts **85**. Once the cushions are positioned in the inserts **85**, pop rivets **91** extend through washers **93** positioned in rivet holes **95** in the cushions to attach the cushions to the top. The pop rivets have their ends extend perpendicularly to their axis, as is well known in the art, to permanently attach the cushions to the top. FIG. **32** identifies preferred embodiments and various aspects of the gap in the top. The top **16** may be one continuous piece with the cushions attached to it by fasteners.

(72) See U.S. patent application Ser. No. 15/802,949, incorporated by reference herein, describing a stepladder that leans upright against an object in a folded position.

(73) Although the invention has been described in detail in the foregoing embodiments for the purpose of illustration, it is to be understood that such detail is solely for that purpose and that variations can be made therein by those skilled in the art without departing from the spirit and scope of the invention except as it may be described by the following claims.

Claims

1. A ladder comprising: a first rail; a second rail; and a top attached to the first rail and second rail, the top including a first side and a second side, the first side including a surface and a gap, the gap including a first rectangular slot and a second rectangular slot extending from the first rectangular slot, the first rectangular slot and the second rectangular slot are configured to receive an external corner of a structure, the surface of the first side including a first notch in the gap such that a first rectangular plank can fit into the first notch, the first notch including a first width, the first side of the top including a first edge that includes a flat surface and angled corners, the angled corners of the top are configured to fit against walls having an internal corner of the structure when the top leans against the internal corner, wherein the top includes a plurality of openings extending through the first side and a pair of discrete and spaced apart cushions configured to contact a surface to avoid marring the surface, and including barbs extending therefrom, wherein the barbs are disposed through the openings in the first side of the top, each discrete cushion of the pair of discrete and spaced apart cushions disposed on and continuously extending along both the flat surface of the first edge and the angled corners of the first edge.

2. The ladder of claim 1 wherein each discrete cushion of the pair of discrete and spaced apart cushions is coupled to the third side using an adhesive.

3. The ladder of claim 1 wherein each discrete cushion of the pair of discrete and spaced apart cushions is coupled to the third side using the barbs and an adhesive.

4. A ladder comprising: a first rail coupled to a second rail via a first plurality of rungs; a third rail coupled to a fourth rail via a second plurality of rungs, the third rail and the fourth rail being connected to the first rail and the second rail via a hinge; a ladder top coupled to at least the first rail and the second rail, wherein the ladder top includes a first side, a second side, a third side, and a fourth side, wherein the third side is coupled to the first rail and fourth side is coupled to the second rail; wherein the first side and the second side are substantially parallel to one another and coupled to the third side and the fourth side of the ladder top, and the ladder top also includes a top surface coupled to the first side, the second side, the third side, and the fourth side, wherein the first side includes a gap with a stepped configuration, a first edge with a flat surface and angled corners, and two discrete and spaced apart pads, wherein the stepped configuration includes a first notch and at least one additional notch on each side of the first notch within the gap, and wherein the two discrete and spaced apart pads are configured to avoid marring of a surface when placed into contact therewith, wherein the first side includes openings extending through the first side and

configured to receive protrusions extending from an edge of the two discrete and spaced apart pads to couple the two discrete and spaced apart pads to the first side, and wherein the two discrete and spaced apart pads are each partly disposed on and continuously extend along both the flat surface of the first edge and the angled corners of the first edge, each of the two discrete and spaced apart pads including: at least one corner protrusion extending from the edge thereof and configured to be received in an opening in the angled corner of the first edge, and at least two surface protrusions extending from the edge thereof and configured to be received in respective openings in the flat surface of the first edge.

5. The ladder of claim 4 wherein the gap with the stepped configuration includes at least three contact surfaces and wherein third and fourth discrete and spaced apart pads are disposed on two of the contact surfaces of the gap with the stepped configuration, wherein the third and fourth discrete and spaced apart pads disposed on the two of the contact surfaces of the gap with the stepped configuration are configured to receive an external corner when the ladder is leaned against the external corner.

6. The ladder of claim 4 wherein the two discrete and spaced apart pads are configured to contact a wall when the ladder is leaned against the wall.

7. The ladder of claim 4 wherein the two discrete and spaced apart pads are configured to contact each side of an internal corner when the ladder is leaned against the internal corner.

8. The ladder of claim 4 wherein the two discrete and spaced apart pads are coupled to the first side via a chemical bond and the at least one corner protrusion and the at least two surface protrusions.

9. The ladder of claim 4 wherein the top surface includes a magnet disposed therein.

10. The ladder of claim 4 wherein the top surface includes tool slots.

11. The ladder of claim 4 wherein the gap with the stepped configuration includes multiple surfaces configured such that the ladder can be leaned against surfaces having various widths, wherein the various widths are narrower than at least one of the multiple surfaces.

12. The ladder of claim 4 wherein the gap with the stepped configuration is configured to receive various non-circular widths of material to aid in stabilizing the ladder against the various non-circular widths.

13. The ladder of claim 4 wherein the angled corners extend between the first side and the third side and the fourth side at approximately a 45 degree angle.

14. The ladder of claim 4 wherein the two discrete and spaced apart pads disposed on at least a portion of the first edge extending from the flat surface to the angled corners span approximately half of a length of each of the angled corners.

15. The ladder of claim 4 wherein the two discrete and spaced apart pads disposed on at least a portion of the first edge extending from the flat surface to the angled corners span approximately a full length of the first side.

16. The ladder of claim 4 wherein the top surface includes an indentation configured to hold tools or materials placed therein.

17. The ladder of claim 4 wherein the gap with the stepped configuration includes two substantially rectangular slots, a first rectangular slot and a second rectangular slot disposed within the first rectangular slot.
